

Tests: Answers

Test 01: Differential Expression Analysis

Goal: This test focuses on the main differential expression analysis workflow in the project.

Students should practice:

- reading DEG result files
- creating new columns from existing columns
- using logical conditions to classify genes
- reproducing the main ideas from the DEG and volcano scripts

Question 1

Inspect the DEG result table.

Tasks:

- print `dim(deg_res)`
- print the first 6 column names
- show the first 3 rows of `deg_res[, 1:5]`

```
dim(deg_res)
```

```
## [1] 17075      9
```

```
names(deg_res)[1:6]
```

```
## [1] "gene_id" "gene_name" "gene_type" "logFC" "AveExpr" "t"
```

```
head(deg_res[, 1:5], 3)
```

```
##               gene_id gene_name      gene_type      logFC
## ENSG00000165197.5  ENSG00000165197.5  VEGFD protein_coding -6.034376
## ENSG00000167434.10 ENSG00000167434.10    CA4 protein_coding -7.012686
## ENSG00000133800.9  ENSG00000133800.9  LYVE1 protein_coding -4.863121
##               AveExpr
## ENSG00000165197.5  -0.5982728
## ENSG00000167434.10 -2.4624963
## ENSG00000133800.9   1.4585257
```

Question 2

Create a new column called `log10FDR`.

Rule:

- `log10FDR = -log10(adj.P.Val)`

```
deg_res$log10FDR <- -log10(deg_res$adj.P.Val)
deg_res$DEG <- "NotSig"
deg_res$DEG[deg_res$logFC > 1 & deg_res$adj.P.Val < 0.05] <- "Up"
deg_res$DEG[deg_res$logFC < -1 & deg_res$adj.P.Val < 0.05] <- "Down"
table(deg_res$DEG)
```

```
##
##   Down NotSig   Up
##   2782 12741 1552
```

Question 3

Keep only significant genes and show the top 5 genes with the largest positive `logFC`.

Tasks:

- create `deg_only` using `DEG != "NotSig"`
- print `dim(deg_only)`
- print the top 5 rows sorted by `desc(logFC)`

```
deg_only <- deg_res %>% filter(DEG != "NotSig")
dim(deg_only)
```

```
## [1] 4334   11
```

```
deg_only %>%
  arrange(desc(logFC)) %>%
  head(5)
```

```
##               gene_id gene_name      gene_type  logFC
## ENSG00000123500.10 ENSG00000123500.10 COL10A1 protein_coding 6.861573
## ENSG00000170373.8  ENSG00000170373.8    CST1 protein_coding 6.586908
## ENSG00000137745.12 ENSG00000137745.12    MMP13 protein_coding 6.332983
## ENSG00000029559.7  ENSG00000029559.7    IBSP protein_coding 5.849066
## ENSG00000099953.10 ENSG00000099953.10    MMP11 protein_coding 5.789660
##               AveExpr      t      P.Value    adj.P.Val      B
## ENSG00000123500.10  5.8002885 16.17469 1.729202e-53 2.001771e-52 110.74879
## ENSG00000170373.8   0.8377648 12.96431 4.082170e-36 2.750713e-35  71.14663
## ENSG00000137745.12  2.7415034 13.68585 8.776159e-40 6.663091e-39  79.47569
## ENSG00000029559.7  -0.2826701 18.40381 5.535171e-67 9.527525e-66 141.58670
## ENSG00000099953.10  7.5370984 21.54809 1.350194e-87 3.730511e-86 188.83396
##               log10FDR DEG
## ENSG00000123500.10 51.69859 Up
## ENSG00000170373.8 34.56055 Up
## ENSG00000137745.12 38.17632 Up
## ENSG00000029559.7 65.02102 Up
## ENSG00000099953.10 85.42823 Up
```

Short Answer

The **DEG** column is a simple label that classifies each gene based on both the direction of change and the significance threshold. **Up** means strongly upregulated in tumor, **Down** means strongly downregulated in tumor, and **NotSig** means the gene does not pass the chosen cutoff.

We classify genes before plotting so the volcano plot can separate biologically meaningful changes from background points. This makes the figure easier to read and helps highlight the most important upregulated and downregulated genes.

Test 02: Enrichment Analysis

Goal: This test focuses on the GSEA workflow in the project.

Students should practice:

- reading GSEA result tables
- extracting important columns
- comparing different GSEA outputs
- explaining what a GSEA result means in simple language

Question 1

Inspect the three GSEA result tables.

Tasks:

- `print dim(gsea_go)`
- `print dim(gsea_kegg)`
- `print dim(hallmark)`
- print the first 3 descriptions from `gsea_go`

```
dim(gsea_go)
```

```
## [1] 10 11
```

```
dim(gsea_kegg)
```

```
## [1] 10 11
```

```
dim(hallmark)
```

```
## [1] 10 11
```

```
head(gsea_go$Description, 3)
```

```
## [1] "mitotic sister chromatid segregation"  
## [2] "sister chromatid segregation"  
## [3] "attachment of spindle microtubules to kinetochore"
```

Question 2

Extract key columns from GSEA GO and GSEA KEGG results.

Tasks:

- from `gsea_go`, keep `Description`, `NES`, `p.adjust`, and `setSize`
- from `gsea_kegg`, sort by `p.adjust` from small to large
- print the first 5 pathway descriptions from the sorted GSEA KEGG table

```
gsea_go_small <- gsea_go %>% select(Description, NES, p.adjust, setSize)
head(gsea_go_small, 5)
```

```
##               Description      NES      p.adjust
## 1      mitotic sister chromatid segregation 3.047349 8.717143e-09
## 2               sister chromatid segregation 2.901035 8.717143e-09
## 3 attachment of spindle microtubules to kinetochore 2.861890 8.717143e-09
## 4               metaphase chromosome alignment 2.853208 8.717143e-09
## 5      mitotic sister chromatid separation 2.844243 8.717143e-09
##      setSize
## 1      190
## 2      229
## 3       54
## 4      101
## 5       64
```

```
gsea_kegg_sorted <- gsea_kegg %>% arrange(p.adjust)
head(gsea_kegg_sorted$Description, 5)
```

```
## [1] "Systemic lupus erythematosus"
## [2] "Cell cycle"
## [3] "Neutrophil extracellular trap formation"
## [4] "Viral carcinogenesis"
## [5] "Alcoholism"
```

Question 3

Use Hallmark GSEA results to compare positive and negative enrichment.

Tasks:

- create `hallmark_positive` using `NES > 0`
- create `hallmark_negative` using `NES < 0`
- print the first 5 descriptions from each table

```
hallmark_positive <- hallmark %>% filter(NES > 0)
hallmark_negative <- hallmark %>% filter(NES < 0)
head(hallmark_positive$Description, 5)
```

```
## [1] "HALLMARK_E2F_TARGETS"      "HALLMARK_G2M_CHECKPOINT"
## [3] "HALLMARK_MYC_TARGETS_V1"   "HALLMARK_MTORC1_SIGNALING"
## [5] "HALLMARK_MITOTIC_SPINDLE"
```

```
head(hallmark_negative$Description, 5)
```

```
## [1] "HALLMARK_ADIPOGENESIS" "HALLMARK_MYOGENESIS"
```

Short Answer

A positive NES usually suggests that genes in that pathway tend to appear near the top of the ranked list, which often indicates relative activation or upregulation in the target condition. A negative NES suggests the pathway is enriched toward the lower end of the ranked list, which often reflects relative suppression.

GSEA results are often sorted by adjusted p value because many pathways are tested at the same time. Ranking by adjusted p value helps prioritize the most statistically reliable pathways after multiple-testing correction.